

AAYUSH RAJ

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Professional Summary

Cybersecurity graduate specializing in **SOC operations, VAPT, threat intelligence, vulnerability management, and security automation**. Hands-on internship experience in risk assessment, alert review, penetration testing, and remediation prioritization. Built Python/FastAPI/React security platforms for **CVE correlation, IOC extraction, CVSS/EPSS/KEV enrichment, RBAC reporting, OSINT workflows, and ML-assisted analysis**. ISC2 Certified in Cybersecurity (CC), ranked in the **Top 1% on TryHackMe**, and seeking entry-level Cybersecurity Analyst, SOC Analyst, VAPT Analyst, or Security Engineer roles.

Technical Skills

Security Domains: SOC Operations, VAPT, Threat Intelligence, Incident Response, Vulnerability Management, Defensive Security, OSINT, Security Automation, Risk Assessment, Blue Teaming

Frameworks and Standards: MITRE ATT&CK, OWASP Top 10, CVSS, EPSS, CISA KEV, NIST Fundamentals, SIEM Workflows

Security Tools: Nmap, Wireshark, Burp Suite, Metasploit, OWASP ZAP, Kali Linux, Splunk, Volatility, Cisco Packet Tracer

Programming and Platforms: Python, PowerShell, SQL, JavaScript, TypeScript, FastAPI, React.js, REST APIs, Docker, Git, Cloud Run, Cloudflare Pages

Data and AI/ML: PostgreSQL, PostGIS, SQLite, Redis, Scikit-learn, TensorFlow, Random Forest, BiLSTM, NLP, Threat Correlation

Professional Experience

Cybersecurity Intern - SOC / VAPT

Dec 2025 – Jan 2026

Invigo Infotech

Hybrid

- Conducted enterprise-level risk assessments across **15+ client systems**, identifying vulnerabilities, misconfigurations, and security gaps.
- Supported SOC workflows including alert review, incident monitoring, security analysis, and remediation prioritization using SIEM-led processes and network security tooling.
- Helped identify **40+ high-risk security weaknesses** and contributed to mitigation planning that improved remediation prioritization efficiency by **30%**.

Offensive Cybersecurity Intern - Penetration Testing

Jun 2025 – Jul 2025

InLighnX Global Pvt Ltd (InLighn Tech)

Remote

- Performed penetration testing and exploit validation on simulated enterprise environments aligned with **MITRE ATT&CK** and **OWASP** practices.
- Used Kali Linux, Burp Suite, Metasploit, and Nmap for reconnaissance, exploitation, and vulnerability analysis, confirming **25+ exploitable attack vectors** across **10+ adversarial simulations**.

Virtual Cybersecurity Intern

Jun 2025 – Aug 2025

Cisco Networking Academy

Remote

- Completed enterprise-focused cybersecurity training covering defensive operations, infrastructure protection, CCNA-level networking, and access control fundamentals.

Selected Security Projects

Agent-less Windows Vulnerability Scanner | *Python, FastAPI, React.js, PowerShell, WinRM, Docker, CVE Mapping* GitHub

- Developed an agent-less Windows security assessment platform for **CVE correlation, posture analysis, compliance checks, remote asset visibility, and SOC-style dashboards**.
- Implemented Windows-native telemetry collection, **RBAC authentication**, risk scoring, and structured reports across **JSON, CSV, HTML, and PDF**.
- Reduced repetitive assessment effort by approximately **60%** through automated evidence collection, report generation, and remediation guidance.

Cyber Threat Intelligence AI System | *Python, FastAPI, React.js, SQLite, Docker, OSINT, Machine Learning*

GitHub

- Built a production-style CTI platform to automate **threat ingestion, IOC extraction, threat correlation, severity scoring, and cyber incident analysis**.
- Integrated OSINT/API-driven feeds, ML-based threat scoring, clustering, analyst dashboards, and JSON artifact generation for security investigation workflows.

OEM Vulnerability Monitoring Tool | *Python, FastAPI, React.js, PostgreSQL, Docker, CVSS, EPSS, KEV*

GitHub

- Built a vulnerability intelligence platform to monitor **10+ OEM security advisory sources** across IT/OT ecosystems and centralize exposure tracking.
- Enriched advisory data with **CVSS, EPSS, KEV, and exploit intelligence**, reducing manual monitoring effort by approximately **70%** through automated classification, reporting, and alerting.

Crime Hotspot Mapping Tool | *Python, FastAPI, React.js, PostgreSQL/PostGIS, DBSCAN, KDE, BiLSTM* GitHub

- Built a geospatial crime analytics platform using **DBSCAN clustering, KDE hotspot detection, and BiLSTM-based FIR classification**.
- Achieved **94.1% classification accuracy** and reduced geospatial query latency by **8.8x** through PostGIS GiST indexing and optimized spatial querying.

Cryptographic Algorithm Identifier | *Python, FastAPI, React.js, Scikit-learn, Entropy Analysis, Random Forest* GitHub

- Developed an AI-powered ciphertext classification platform using entropy analysis and Random Forest models to identify likely cryptographic algorithms.
- Built REST APIs for real-time prediction and practical security analysis workflows.

Education

Government Engineering College, West Champaran

B.Tech in Computer Science and Engineering (Cyber Security), CGPA: 8.23 / 10

Nov 2022 – May 2026

Bihar Engineering University

Certifications, Labs and Achievements

- **Certifications:** ISC2 Certified in Cybersecurity (CC); Cisco Ethical Hacker; Cisco Cybersecurity Essentials; Cisco Introduction to Cybersecurity; CCNA 1, CCNA 2, CCNA 3; Networking Essentials; NPTEL Ethical Hacking; Advanced Ethical Hacking Bootcamp.
- **Job Simulations and Labs:** TATA Cybersecurity Analyst, Mastercard Cybersecurity, ANZ Cyber Security Management, Cisco Cybersecurity Virtual Internship, TryHackMe Top 1% learner with hands-on labs in web security, network security, privilege escalation, VAPT, and SOC operations.
- **Research and Activities:** Published national-conference research on DBSCAN-KDE crime hotspot mapping; won college-level badminton, cricket, and volleyball tournaments; participated in CTF-style attack-defense simulations and technical presentations.